

Premnaath Varadharajan

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EDUCATION

University of Southern California, Los Angeles, CA, USA

Jun. 2025 – Present

Master of Science in Computer Science

GPA: 3.89/4.0

Anna University – College of Engineering, Guindy, India

Aug. 2019 – Apr. 2023

Bachelor of Technology in Information Technology

CGPA: 9.07/10.0

EXPERIENCE

Amazon Web Services

May 2026 – Aug. 2026

Software Development Engineer Intern

East Palo Alto, CA

- Architected a write-through memcache layer for Amazon Athena's highest-volume read API (GetQueryExecution) using Java and Spring, reducing DynamoDB Read Capacity Units (RCU) consumption by over 50%.
- Engineered an AWS Lambda cache invalidator triggered by DynamoDB Streams to maintain data freshness across all upstream writers, requiring zero code modifications to external packages.
- Implemented a token refund mechanism within the Coral service chain, effectively quadrupling customer API throughput for repeat reads without increasing per-account throttle limits.
- Deployed the end-to-end architecture using AWS CDK across 37 gamma regions and 1 production region, configuring CloudWatch EMF metrics and a six alarm composite for system observability.

Oracle

Jan. 2023 – Mar. 2025

Associate Software Developer

Hyderabad, India

- Shipped a complete integration of Oracle's proprietary DataGrid component with Microsoft Excel, creating a reusable library that streamlined development for other engineers (AXCellence Award).
- Optimized performance of Aconex suite modules by using thread pools for concurrency and progressive loading for lazy UI, cutting load times by 27% and making workflows more efficient.
- Led customer meetings alongside PM to define use cases and convert them into prioritized, shippable features.

Project Intern

- Developed a panning canvas UI component from scratch to enable navigation across screens using mouse drag.
- Handled DevOps tasks, including containerizing applications and creating deployment pipelines for a new product.

PROJECTS

Linearizable Distributed KV Store | C++, gRPC, Raft Consensus, Protocol Buffers

2026

- Implemented Raft consensus across independent processes with randomized leader election, log replication, majority-based commits, and recovery from network partitions.
- Built a linearizable Put/Get/Append service with leader aware retries, duplicate suppression, quorum confirmed reads, and concurrent performance benchmarks.

Linear Space Sequence Aligner | C++, Dynamic Programming, Divide and Conquer

2025

- Developed a sequence aligner using Hirschberg's algorithm to reduce space complexity from $O(mn)$ to $O(\min(m, n))$.
- Integrated performance metric tracking to quantitatively validate significant efficiency gains in memory usage.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, PostgreSQL, Lua

Developer Tools: Git, Linux, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Shell Scripting

Libraries/Frameworks: Flask, Django, Supabase, React, pytest, gRPC, C++ STL

Machine Learning: RAG, Embeddings, LangChain; IR/NLP integrations

Concepts: Distributed Systems, Operating Systems, DBMS, Computer Networks, Caching, Logging, Multithreading

NOTABLE HIGHLIGHTS

- Received the AXCellence Award (Extra Mile category) from CEGBU's Vice President for FY24 Q3 at Oracle.
- Won a \$150 cash prize as runner-up in Oracle's IGIU FY24 Hackathon.
- Participated in Apple's WWDC Swift Student Challenge 2023; earned a free entry ticket worth \$1000 to Apple Park, CA.
- Awarded Certificate of Merit by CBSE for securing a perfect 10 CGPA in AISSE (2015-2016).
- Contributed to [TLDR](https://tldr.sh) pages, an open-source community-driven command line tool.
- Online Profiles: [LeetCode](https://leetcode.com/premnaath), [CodeChef](https://www.codechef.com/premnaath), [CodeForces](https://www.codeforces.com/premnaath), [GeeksforGeeks](https://www.geeksforgeeks.org/premnaath), [HackerRank](https://www.hackerrank.com/premnaath), [Stack Overflow](https://stackoverflow.com/premnaath)